

Requirement Management Workflow

On-Demand Model for NAPCO Nucleus (NN)

Owner	Assad Zaman	Date	7 May 2026
System	NAPCO Nucleus (NN)	Status	DRAFT FOR REVIEW

1. Purpose

Client requirements arrive across fragmented channels — Microsoft Teams calls, Teams chats and DMs, email, and Google Drive. Consolidating them into auditable documents is presently a manual, time-consuming task. This workflow puts **NAPCO Nucleus** behind that capture, while keeping every outbound action under direct human control.

2. Operating Principles

Four non-negotiable principles govern this workflow:

- **User-triggered.** NAPCO Nucleus performs work only on explicit command. There is no scheduled background polling of any channel.
- **Human-in-the-loop.** Every requirement document is reviewed by Assad before any email draft is composed.
- **No automated outbound email.** NAPCO Nucleus prepares the draft and the attachment. Assad sends the email himself, from his own mail client.
- **Speaker attribution preserved on calls.** Assad's voice and the other party's voice are recorded on separate tracks so summaries cannot misattribute who said what.

3. Input Channels

Channel	Mechanism	What it captures
Teams audio calls	Local recording — microphone + system loopback	Dual-track audio; transcribed and translated to English (faster-whisper large-v3)
Teams messages	Reads Teams desktop's local message cache	Group chats and DMs, queryable by chat name and time range
Email	IMAP poll of allowlisted senders	Message bodies, plus PDF / Word .docx / plain text attachments extracted to text

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Google Drive	Drive API + content extractors	Audio (Whisper), PDF (pypdf), Word .docx (python-docx), plain text

4. Core Workflow Steps

- 1 **Capture.** Assad triggers a read or recording for the channel he wants.
- 2 **Process.** NAPCO Nucleus handles transcription, translation, or parsing as the channel requires.
- 3 **Output.** A structured Microsoft Word document is produced for Assad's review.
- 4 **Final action.** On request, NAPCO Nucleus composes a draft email with the document attached. Assad reviews and sends it manually.

5. Technical Foundation

The workflow reuses components already present in the *napco-labs* codebase. No new external services and no new self-hosted runners are required.

- **Teams Requirement Watcher (TRW)** — existing call-recording and call-transcription pipeline. Bangla → English translation is built in.
- **NAPCO Nucleus core** — existing Word document writers, IMAP email reader, and Google Drive content ingest.

6. Items Pending Approval

Before implementation continues, three items need confirmation:

- 1 The proposed on-demand operational model.
- 2 The designated Google Drive folder for audio archival.
- 3 Confirmation that direct OpenProject integration is **not** required at this stage.